sector_balanced

equity strategy | family: equity-neutralized | status: validated | derived from `conviction_tilt`

What this strategy does

Conviction Tilt, also balanced across 12 industry groups; conservative 1.5x borrowing limit

Why it should work. Same drift, with sector and size risk diversified away so only the earnings signal is left.

Design notes (from the strategy's source file).

Sector Balanced -- Conviction Tilt plus an equal budget for each industry group.

A diagnostic found Conviction Tilt's book putting up to 29% of gross weight in one FF12 sector in some quarters (8.3% if spread evenly): an unpriced bet unrelated to earnings drift. Neutralizing each (side, size, sector) cell to an equal budget removes it. Best Sharpe / shallowest drawdown of the hand-built strategies at its conservative 1.5x leverage cap.

parameter	value
base_unit_fraction	0.0015
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

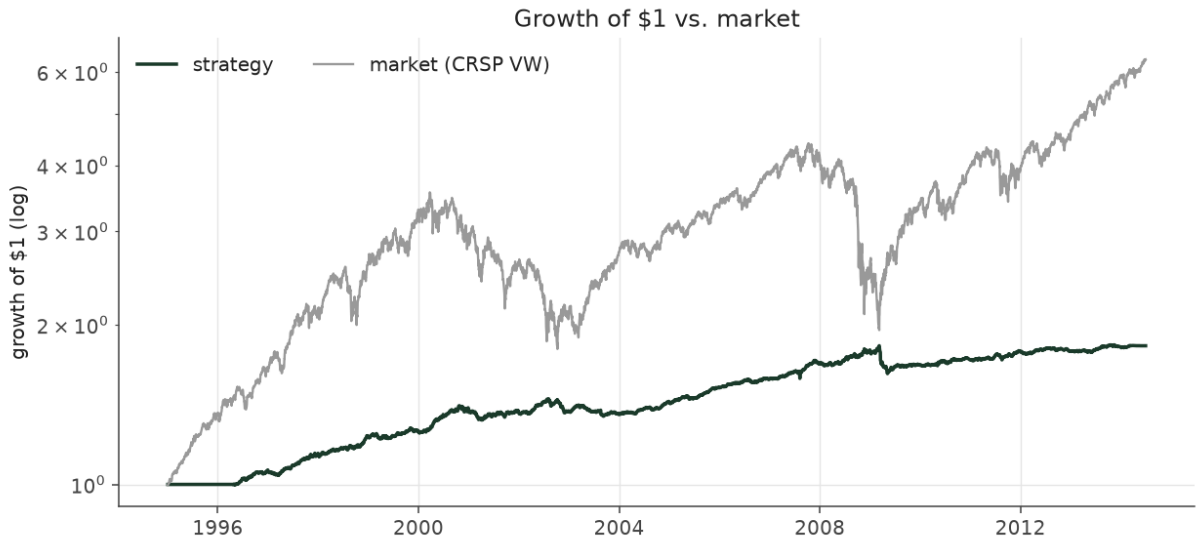
metric	sector balanced	market (CRSP VW)
Annualized return	3.14%	9.94%
Annualized volatility	2.89%	19.35%
Sharpe (NAV-pct)	1.08	0.59
Sortino	1.52	
Max drawdown	-11.4%	-55.5%
Calmar	0.28	
Final NAV ($10M start)	$18,260,949	
Worst day	-3.27%	
% positive days	50.8%	
Sharpe 1996-2001	1.72	
Sharpe 2002-2007	1.25	
Sharpe 2008-2013	0.41	

Correlation of daily returns with the market: **-0.00**.

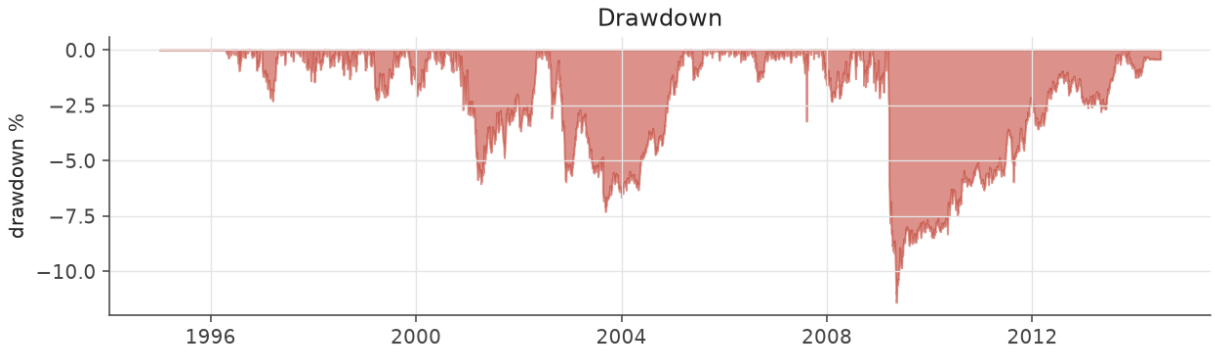
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.08

• best return among equity: `market_boost_max` (0.116); this one: 0.0314

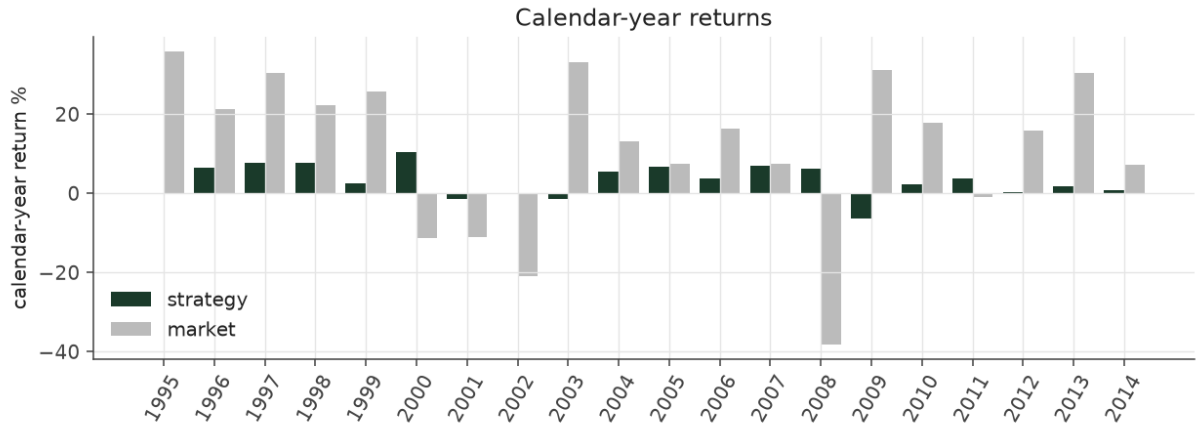
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.275



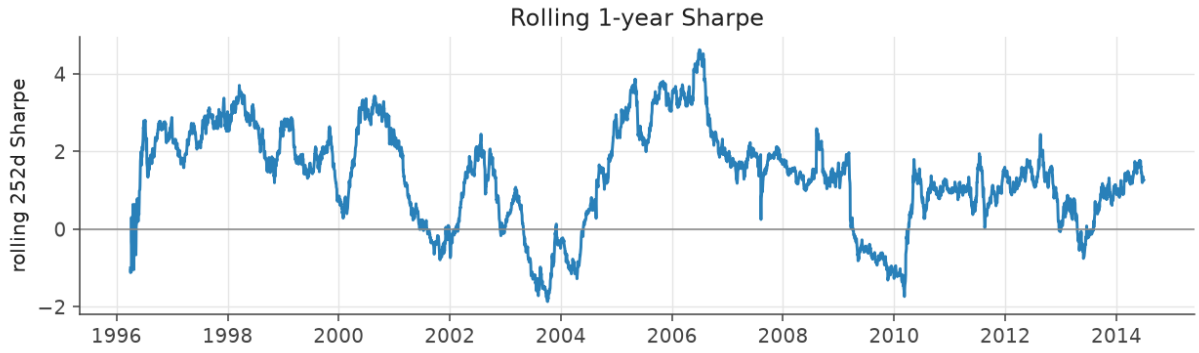
NAV (log scale)



Drawdown from running peak

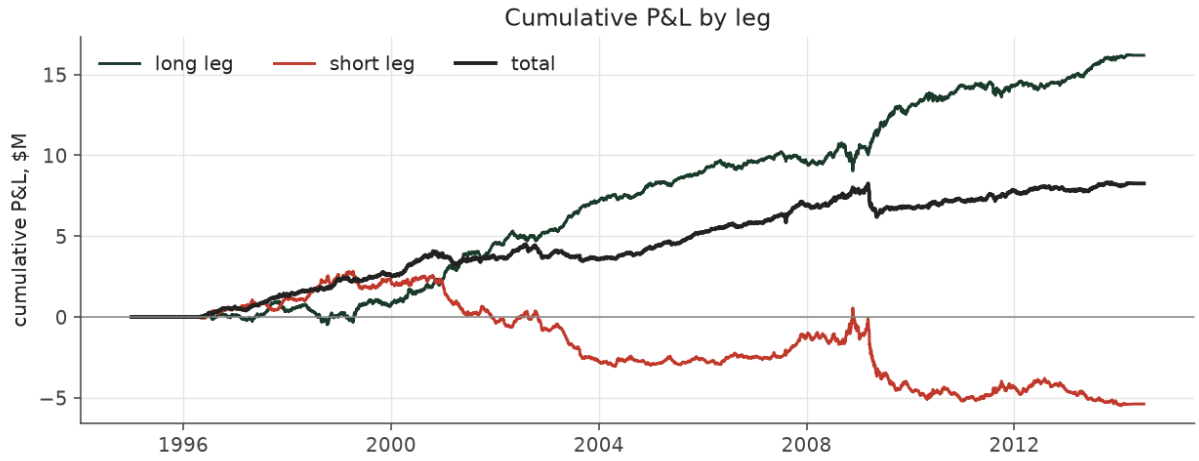


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	16.19	-5.39
P&L after costs ($M)	14.95	-6.69
of which common (average-event) component ($M)	8.99	-9.21
of which excess / PEAD component ($M)	7.20	3.82
excess minus costs ($M)	5.96	2.51
gross return per $ traded (bp)	277.94	-90.79
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.26	-0.36

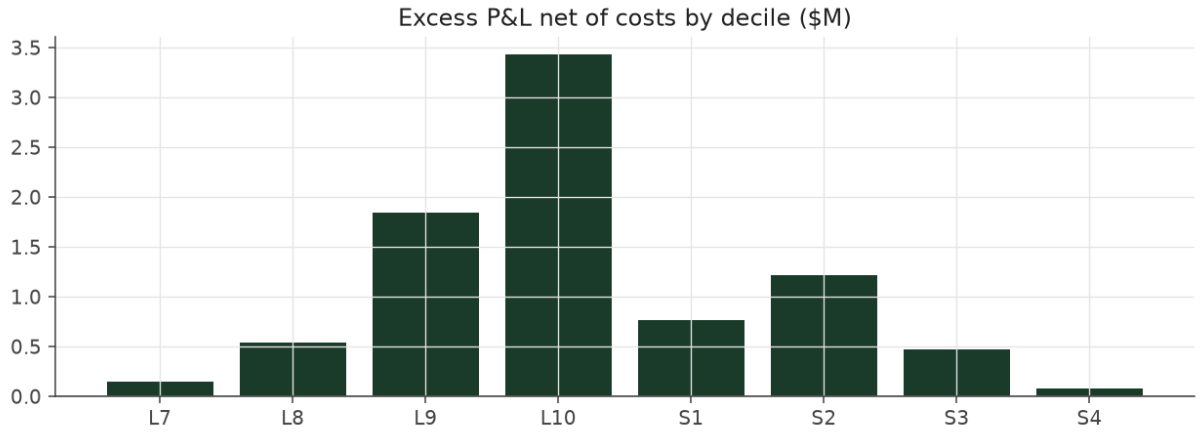
Daily P&L correlation between the two legs: **-0.79**.

• **The short leg loses money in absolute terms** (-5.4M before costs) while the long leg earns +16.2M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 9.0M / 9.2M. Stripping that common component, the surprise signal itself earned +7.2M on the long side and +3.8M on the short side.

• The two legs' daily P&L are -0.79 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.26 long / -0.36 short vs 1.08 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	6,082	37.83	0.76	0.23	0.08	0.14	1.7%	38.00
long	8	16,057	127.06	2.81	0.81	0.27	0.54	6.4%	42.58
long	9	18,361	179.68	4.19	2.23	0.38	1.85	21.8%	102.81
long	10	20,072	238.05	7.19	3.94	0.51	3.43	40.5%	144.16
short	1	18,251	239.38	-3.19	1.31	0.55	0.76	9.0%	31.85
short	2	18,738	203.97	-1.97	1.65	0.44	1.21	14.3%	59.32
short	3	14,800	133.12	-1.39	0.75	0.28	0.47	5.5%	35.11
short	4	2,262	16.74	-0.14	0.11	0.03	0.07	0.9%	44.07



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,854	150.43	1.21	2.08	0.60	1.48	17.4%	98.20
2	21,876	234.74	2.57	3.32	0.68	2.64	31.2%	112.65
3	24,597	259.69	2.16	2.71	0.56	2.15	25.4%	82.86
4	26,557	265.14	1.43	1.79	0.42	1.37	16.2%	51.79
5	27,739	265.83	0.89	1.12	0.29	0.83	9.8%	31.26

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,187	119.70	1.96	2.75	0.60	2.15	25.4%	179.53
2	22,184	257.13	2.03	3.06	0.77	2.29	27.1%	89.20
3	24,828	271.79	1.83	2.44	0.54	1.89	22.3%	69.63
4	26,579	265.11	1.61	1.83	0.37	1.46	17.2%	55.13

5	27,841	262.03	0.82	0.94	0.26	0.68	8.0%	25.79

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	40,606	393.78	0.32	3.08	0.80	2.29	27.0%	58.04
41-60d	27,196	297.47	1.18	2.56	0.69	1.86	22.0%	62.59
61-90d	37,427	381.34	3.58	2.62	0.75	1.87	22.1%	49.12
>90d	9,394	103.24	3.18	2.77	0.31	2.46	29.0%	237.99

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,241	304.99	3.68	4.40	0.65	3.75	44.3%	123.05
2002-07	37,882	393.27	3.11	3.98	0.83	3.14	37.1%	79.95
2008-13	44,500	477.57	1.47	2.65	1.07	1.58	18.6%	33.09

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	20,671	108.71	0.44	0.67	0.24	0.43	5.1%	39.70
Chems	2,553	82.04	0.28	0.48	0.16	0.32	3.8%	39.15
Durbl	2,829	90.23	0.96	1.18	0.19	1.00	11.7%	110.33
Enrgy	6,341	98.29	1.18	1.39	0.20	1.19	14.0%	120.86
Hlth	11,339	108.11	0.50	0.75	0.24	0.51	6.0%	47.18
Manuf	11,787	103.70	0.70	0.97	0.23	0.74	8.7%	71.03
Money	16,516	99.52	1.01	1.28	0.24	1.03	12.2%	103.99
NoDur	4,907	97.23	0.47	0.64	0.21	0.43	5.1%	44.73
Other	17,460	106.30	0.99	1.26	0.25	1.01	11.9%	95.03
Shops	11,520	106.61	1.26	1.50	0.23	1.27	15.0%	119.04
Telcm	3,960	98.15	0.70	0.91	0.20	0.71	8.3%	72.10
Utils	4,740	76.93	-0.23	-0.01	0.15	-0.16	-1.9%	-21.11

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	38,373	366.60	0.83	1.93	0.71	1.22	14.4%	33.27
low vol	38,203	409.42	3.77	4.83	0.98	3.85	45.4%	93.98
mid vol	36,279	376.16	3.32	3.89	0.80	3.09	36.5%	82.15

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	3.14%	-0.00%
0.5%	0.6	2.96%	0.18%
1.0%	1.2	2.78%	0.36%
2.0%	2.4	2.39%	0.75%
5.0%	6.1	1.02%	2.12%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.